EchoTrace: Do Language Models Mistake Repeated Reports for Independent Corroboration?

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Abstract

When several documents support a claim, a research assistant may treat them as independent confirmation even if they all derive from one observation. We study this *false corroboration* in language models. We release EchoBench-Synthetic v1.4, a chronology-valid, gold-by-construction benchmark of 3,960 cases from 360 world-disjoint micro-worlds, 36 topics, and six prose styles, two of which appear only in held-out worlds. In an exploratory study on 12 validation worlds, three freely accessible models (Qwen3.8-27B, GPT-OSS-120B, and Gemini 3.5 Flash-Lite) answer under four conditions: no guidance, a dependence reminder, source families proposed by EchoGraph, an interpretable CPU detector, and gold families. Models must report how many independent sources the evidence contains, which lets us score overcounting automatically. Without guidance, GPT-OSS overcounts dependent sources in 0.467 of cases (95% CI [0.273, 0.667]), usually by reporting the number of documents; the other two models rarely do. Guidance removes overcounting but over-corrects: GPT-OSS then undercounts genuinely independent sources in 0.667 of cases with EchoGraph families and 0.722 even with gold families, and Gemini abstains on many determinate cases. No model's confidence in its verdict depends on how many independent sources support it. These are development findings, not confirmatory tests; we freeze the held-out protocol that would test them. EchoGraph itself reaches family-pair F1 0.920 in a single held-out run after the freeze but remains weak on shared datasets. All experiments run on CPU and free API tiers; no human validation is claimed.

1 Introduction

A research assistant may cite four pages and describe a finding as repeatedly confirmed. If three pages summarize the first, the answer contains four documents but one informational origin. We call this *false corroboration*: apparent support increases because dependent reports are counted as independent observations. Citation checks do not expose it, because every page genuinely states the claim.

Source dependence has long mattered in data integration (Dong et al., 2009), and citation networks can amplify claims through repeated reference (Greenberg, 2009). Current evaluations of attributed and deep-research systems measure correctness, citation support, evidential force, and synthesis quality (Gao et al., 2023; Du et al., 2025; Qian et al., 2026). They do not directly measure whether a system knows how many *distinct* observations its sources represent. Recent formal and empirical work shows that aggregating multiplied agent reports can destroy calibration when their shared ancestry is ignored (Bara, 2026). Whether language models make this error when provenance must be read from the documents themselves, and whether simple provenance metadata prevents it, has not been tested directly.

We test it in a controlled setting where provenance is known by construction. Each case gives a model the same four claim-relevant documents, varying only whether they share an origin. We ask three questions:

1. **False corroboration.** Without guidance, do models report more independent sources than exist when documents share an origin?

2. **Confidence sensitivity.** Holding the claim and verdict fixed, is a model more confident when evidence comes from four independent sources than from one?

3. **Mitigation.** Do source families proposed by an interpretable detector reduce false corroboration, beyond a plain reminder, without lowering verdict accuracy or causing models to

undercount independent evidence?

Our contributions are:

- a zero-cost exploratory evaluation of three model families across four guidance conditions, with automatically scored source counts and whole-world bootstrap intervals, showing that guidance can trade overcounting for undercounting;
- a frozen held-out confirmatory protocol, amended from these development observations with an over-correction guard, that others can run under the same free quotas;
- EchoBench-Synthetic v1.4, with chronology-valid lineage, observable provenance cues, 36 topics, and held-out prose styles; and
- EchoGraph, a CPU-only logistic source-family detector, evaluated against document, domain, lexical, TF–IDF, and sentence-embedding baselines, together with an account of a construction defect in our earlier release and its consequences.

The study measures how models use *visible* provenance in synthetic documents. It does not establish open-web provenance recovery, factual truth, or human-validated answer quality.

2 Related Work

Attribution and research-agent evaluation. ALCE evaluates correctness and citation quality in attributed answers (Gao et al., 2023). Deep-research benchmarks evaluate long reports and cross-source reasoning (Du et al., 2025; Xie et al., 2026), evidence-force evaluation separates relevance from warrant (Qian et al., 2026), and audits show that displayed citations need not provide verified attribution (Onweller et al., 2026). Source choice can also change answers in multi-source medical RAG (Li et al., 2026). EchoTrace measures a different property: relationships *between* supporting sources.

Dependence and diversity. Copy-detection models for data integration avoid double-counting repeated values (Dong et al., 2009). MMR reduces redundancy by trading relevance against novelty (Goldstein and Carbonell, 1998), but textual diversity is not evidential independence: independent studies may use similar language and derivatives may paraphrase. LineageRAG builds evidence lineages grounded in verbatim source spans (Zheng et al., 2026); EchoTrace instead infers which documents share an origin.

Report multiplicity. Bara (2026) formalizes when an additional report adds no evidence, and shows with controlled LLM-agent extraction on synthetic documents that naive aggregation loses calibration as reports of one evidence root multiply. It also finds that report similarity, not true ancestry, drives a similarity-based deduplicator, which parallels our same-wording condition. That work studies aggregation over agent reports. EchoTrace is complementary and narrower: it asks whether a single model, reading the documents themselves, counts dependent reports as independent sources, and whether provenance guidance changes that count in either direction.

3 Task and Measures

For a focal claim c, let $\mathcal{D}_c = \{d_1, \ldots, d_n\}$ be its relevant documents. In a determinate case, a latent mapping $z(d)$ assigns each document to one source family, giving

$$\mathcal{F}_c = \{F_k\}_{k=1}^{K}, \quad F_k = \{d \in \mathcal{D}_c : z(d) = k\}, \tag{1}$$

where K counts informational origins, not URLs or publishers. In an indeterminate case the correct response is to abstain from a count.

Language-model measures. A model returns a verdict (supported, contradicted, insufficient), verdict probabilities, citations, and $\widehat{K}$, the number of independent sources bearing on the claim, or `null` if this cannot be determined. The *false-corroboration rate* (FCR) is the fraction of responses on dependent cases ($K < 4$) with $\widehat{K} > K$; an abstention is not an overcount. *Confidence sensitivity* is the within-world difference in the probability assigned to the gold verdict between the four-independent and single-origin versions of the same claim; a model that weighs independence should give a positive value. We also report undercounting on independent cases ($\widehat{K} < K$), abstention on determinate and on indeterminate cases, verdict accuracy, and the share of gold-verdict probabilities at a 0.99 ceiling, which shows whether confidence could rise at all.

Detector measures. For detectors returning a partition $\widehat{\mathcal{F}}_c$, we report root-count mean absolute

Table 1: Provenance conditions. K is the gold number of source families among the four relevant documents; "–" denotes an abstention target. $\star$ marks conditions used in the language-model study.

Condition	K	Intended distinction
Single origin*	1	root with three derivatives
Verbatim derivative	1	republication with visible source
Attributed paraphrase	1	visible source links
Two roots*	2	two root–summary pairs
Shared dataset*	3	two analyses reuse one dataset
Study and replications	3	summary plus two replications
Four independent*	4	four separate samples
Same wording independent*	4	wording without dependence
Same domain independent	4	publisher without dependence
Unattributed repetition	–	repetition, provenance unknown
Ambiguous provenance*	–	unknown; abstain

error $\frac{1}{N}\sum_i |\widehat{K}_i - K_i|$ and exclusive-family pair F1 over same-family document pairs.

Source-family dependence is not truth or reliability. Two independent studies may share a bias, and a derivative can accurately summarize strong evidence.

4 EchoBench-Synthetic v1.4

Construction. EchoBench contains 360 fictional micro-worlds, each with one quantitative claim and one verdict, crossed with 11 provenance conditions (Table 1). Every case has 12 documents: four claim-relevant documents and eight distractors. Whole worlds are assigned to train, validation, and test (216/72/72), with verdicts balanced within each split. Documents use reserved `.example` domains and opaque identifiers. Public files contain only queries and documents; evaluator files contain relevance, verdicts, lineage, families, conditions, and splits.

Changes from earlier releases. Version 1.4 addresses three problems found in our earlier releases. First, *chronology*: an audit of the v1.1 training split found 577 of 2,250 lineage edges whose source was dated after its derivative (Section 7.6). Generation now moves each derivative after its sources; v1.4 has zero reversed edges. Second, *observability*: a diagnostic review found that two conditions asserted lineage that the public text did not reveal. Verbatim republications now carry a visible upstream URL and attribution, and unattributed repetition now has an abstention target rather than a hidden chain. Third, *templating*: v1.4 has 36 topics (three times v1.1), six prose styles per provenance role, and varied distractors. Two styles (forum and memo) appear only in half of the test worlds, so part of the test set uses unseen surface forms.

Checks. Structural validation confirms counts, unique identifiers, verdict balance, public/gold separation, split isolation, and chronology. A nearest-centroid probe on four coarse metadata features (mean year, domain count, URL path shape, identifier length) predicts the 11 conditions with accuracy 0.182 (chance 0.091; prespecified failure threshold 0.55). This probe cannot rule out textual shortcuts. Labels are deterministic consequences of the generator, not human judgments.

5 Language-Model Study

Design. We use the six starred conditions in Table 1 for the 12 validation worlds (four per verdict): 72 cases per arm. No test-split world has been sent to any model. Each request contains the same four relevant documents in their public order, with title, URL, date, outbound links, and text. Four arms differ only in their guidance: *standard* (none); *reminder* (repeated reporting need not add independent evidence, similar wording alone does not establish dependence, acknowledge unclear provenance); *EchoGraph* (reminder plus detector-proposed families, labelled as possibly wrong); and *oracle* (identical wording, gold families; for indeterminate cases, a statement that provenance could not be determined). The oracle arm measures the value of correct metadata; it is not a deployable method. Arm order rotates across cases.

Every arm asks for $\widehat{K}$. This primes all arms equally toward considering independence, so the standard arm is a conservative baseline: it measures counting errors that remain even when the model is explicitly asked to count.

Models. We select three model families with verified free access on the study date: Qwen3.8-27B and GPT-OSS-120B (low reasoning effort) through Groq, and Gemini 3.5 Flash-Lite through Google's API. Decoding uses temperature 0 and at most 2,048 output tokens. Requested and returned model identifiers are logged. Requests rejected before generation (rate limit or unavailability) are re-

tried with back-off; completed responses, including invalid ones, are never re-requested. Invalid outputs count against coverage.

Coverage. Gemini answered all 288 planned requests. The two Groq models were stopped by the free tier's daily limit of 200,000 tokens per model (about 190 of our requests): Qwen and GPT-OSS each completed all 24 requests for 8 of the 12 worlds, for coverage of 0.708 and 0.764 of planned requests in their lowest arm. Missing requests are the last worlds in plan order, not selected by outcome. Three GPT-OSS responses cited nonexistent documents and count as invalid.

Analysis. The resampling unit is the micro-world. We report 95% percentile intervals from 10,000 whole-world bootstrap replicates, with contrasts paired by case. With 8–12 worlds per model, the intervals are wide and every result is exploratory: these worlds were also used to revise the protocol (Section 7.5), so they cannot test hypotheses formed from them. We apply no multiplicity correction and claim no confirmed effect.

6 EchoGraph Detector

For each time-ordered pair of documents, EchoGraph computes direct-link presence, attribution cues, token and five-gram Jaccard similarity, bag-of-words cosine, numeric overlap, same-domain status, and clipped time separation. A logistic classifier is trained on all pairs from determinate training cases to predict whether the two documents share a family. Pairs above a threshold are joined, and connected components are the families. The threshold (0.40) is chosen on validation by lowest root-count MAE. Baselines count URLs, assign all documents to one family, group by domain, or join pairs above a bag-of-words, TF–IDF, or MiniLM (`all-MiniLM-L6-v2`, pinned revision) cosine threshold. No method uses a language model or GPU.

7 Results

All results in this section are from validation worlds and are exploratory.

7.1 False corroboration (RQ1)

Without guidance, GPT-OSS overcounts on 0.467 of dependent cases [0.273, 0.667]. The error has a consistent form: in 12 of its 14 overcounts the model reported four independent sources, the number of documents. It did the same on 7 of 9 ambiguous cases, where the correct response is to abstain. Qwen (0.037) and Gemini (0.056) rarely overcount, and both intervals include zero. Qwen nevertheless reported four sources on 3 of 8 ambiguous cases. False corroboration therefore appears in one of three models when provenance is visible, and it takes the form of counting documents.

7.2 Confidence sensitivity (RQ2)

Confidence sensitivity is indistinguishable from zero for every model and arm: a model is no more confident in the gold verdict when four independent sources support it than when one origin does. For Qwen and Gemini this is saturation: 0.896 and 0.979 of their gold-verdict probabilities are at least 0.99, so they could not increase. GPT-OSS stays below the ceiling (0.312) yet still shows no difference (0.004 [-0.005, 0.013]). In this setting, stated verdict probabilities express how clearly the documents state the claim, not how much independent evidence supports it.

7.3 Guidance and over-correction (RQ3)

Every form of guidance eliminates GPT-OSS's overcounting (Table 3), but the plain reminder does so as completely as EchoGraph families do (EchoGraph − reminder: 0.000). The reduction comes with a large increase in undercounting (0.667 [0.375, 0.929]): with guidance, GPT-OSS reports a single source for most cases, including independent ones. This persists in the oracle arm, where the prompt lists the four correct singleton families: on 13 of 18 independent cases GPT-OSS still reported one source (undercount 0.722). The model does not appear to use the metadata; instead, the instruction to consider dependence seems to shift its default answer from "count the documents" to "one origin". Gemini over-corrects differently. The counts it gives remain accurate, but with guidance it abstains on determinate cases (0.217 with EchoGraph versus 0.033 without), mostly on independent ones: 9 of its 13 abstentions in the EchoGraph arm were on four-independent or same-wording cases. The reminder also lowered Gemini's verdict accuracy (-0.069). Qwen counted almost perfectly in every arm, leaving nothing for guidance to fix.

Our protocol's original mitigation criterion checked only verdict accuracy. By that criterion,

Table 2: Development language-model results on validation worlds. n: schema-valid responses per arm. FCR: false-corroboration rate on dependent cases (lower is better). Sens.: confidence sensitivity, $P(\text{gold} \mid K{=}4) - P(\text{gold} \mid K{=}1)$ within world (positive is appropriate). Ceil.: share of gold-verdict probabilities ≥ 0.99. Under: undercount rate on independent cases. Null_K: abstention on determinate cases. Abst.: abstention on ambiguous cases. Intervals: 10,000-replicate whole-world bootstrap.

Model	Arm	n	FCR [95% CI]	Sens. [95% CI]	Ceil.	Under	Null_K	Abst.	Acc.
Qwen3.8-27B	Standard	51	0.037 [0.000, 0.121]	0.006 [0.000, 0.013]	0.896	0.000	0.000	0.625	0.902
	Reminder	51	0.037 [0.000, 0.121]	0.006 [0.000, 0.013]	0.438	0.062	0.023	0.750	0.882
	EchoGraph	51	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.542	0.000	0.047	0.375	0.863
	Oracle	52	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.521	0.059	0.023	0.750	0.865
GPT-OSS-120B	Standard	57	0.467 [0.273, 0.667]	0.004 [-0.005, 0.013]	0.312	0.000	0.021	0.222	0.877
	Reminder	58	0.000 [0.000, 0.000]	-0.034 [-0.092, 0.020]	0.021	0.526	0.265	0.667	0.776
	EchoGraph	56	0.000 [0.000, 0.000]	-0.048 [-0.104, 0.012]	0.104	0.667	0.170	0.333	0.804
	Oracle	55	0.000 [0.000, 0.000]	-0.003 [-0.038, 0.033]	0.043	0.722	0.109	0.556	0.818
Gemini 3.5 Flash-Lite	Standard	72	0.056 [0.000, 0.139]	0.000 [0.000, 0.000]	0.979	0.000	0.033	0.833	0.708
	Reminder	72	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.896	0.000	0.250	1.000	0.639
	EchoGraph	72	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.938	0.000	0.217	1.000	0.722
	Oracle	72	0.000 [0.000, 0.000]	0.006 [0.000, 0.013]	0.896	0.000	0.267	1.000	0.694

Table 3: Paired development contrasts (EchoGraph − standard, and secondary contrasts). Negative ΔFCR means fewer overcounts; positive ΔUnder means more undercounts (worse). Intervals: 95% whole-world bootstrap.

	Qwen	GPT-OSS	Gemini
EchoGraph − standard			
ΔFCR	-0.037	-0.448	-0.056
95% CI	[-0.121, 0.000]	[-0.639, -0.259]	[-0.139, 0.000]
ΔUnder	0.000	0.667	0.000
95% CI	[0.000, 0.000]	[0.375, 0.929]	[0.000, 0.000]
ΔAcc	-0.039	-0.071	0.014
95% CI	[-0.093, 0.000]	[-0.152, 0.000]	[-0.069, 0.125]
Secondary ΔFCR			
Reminder − std.	0.000	-0.467	-0.056
EchoGraph − rem.	-0.037	0.000	0.000
Oracle − EchoGraph	0.000	0.000	0.000
Coverage (min. arm)	0.708	0.764	1.000

GPT-OSS would have shown a large, apparently successful reduction in false corroboration. A mitigation for overcounting must therefore be evaluated together with undercounting and abstention on independent evidence.

7.4 Detector results

On validation, EchoGraph reaches pair F1 0.914 and MAE 0.085, against at most 0.452 F1 for any baseline (Table 4). Removing link features lowers F1 to 0.698: once provenance is made observable, much of the task is reading explicit links and attribution. Shared datasets remain the main failure (F1 0.325), because a reused dataset is signalled by an identifier rather than by a link between documents. The single held-out run after the protocol freeze agrees with validation: EchoGraph reaches F1 0.920 [0.907, 0.933] and MAE 0.086, and shared datasets remain the failure (F1 0.369). High detector accuracy is a property of this construction, not evidence of open-web provenance recovery.

Table 4: Source-family detection on v1.4 determinate cases. Validation selected the thresholds; test is a single run after the protocol freeze (648 cases, 72 worlds). Higher F1 and lower MAE are better.

	Validation		Test	
Method	F1	MAE	F1	MAE
URL singletons	0.333	1.444	0.333	1.444
All one family	0.452	1.556	0.452	1.556
Domain	0.222	1.778	0.222	1.778
Bag of words	0.258	1.370	0.276	1.377
TF–IDF	0.369	1.309	0.376	1.315
MiniLM	0.450	1.552	0.448	1.546
EchoGraph	**0.914**	**0.085**	**0.920**	**0.086**
no links	0.698	0.520	0.697	0.509

7.5 From development to a confirmatory test

These observations changed the held-out protocol, which was then frozen by SHA-256 before any test-split access. Mitigation now requires, in addition to lower FCR and verdict accuracy within 0.05 of the standard arm, that undercounting on independent cases rise by no more than 0.10; models whose standard-arm FCR is zero are reported as untestable rather than as failures; and confidence sensitivity is reported with the ceiling rate. The amended protocol covers all 72 test worlds (1,728 requests per model). Under the free daily token limit, it requires about ten days per Groq model, which is why this paper reports development results only.

7.6 What the v1.1 chronology defect taught us

Our earlier release (v1.1; 2,640 cases, 240 worlds) was evaluated once under a frozen protocol. EchoGraph obtained held-out pair F1 0.734 [0.704, 0.762] and MAE 0.455, against 0.300 for URL counting. A training-only audit after the freeze then found that 577 of 2,250 lineage edges (26%) ran backwards in time. Because EchoGraph orders document pairs by date before computing link features, the defect could affect the detector itself, not only its chronology feature, so those scores cannot be carried over to corrected data. We preserved the frozen v1.1 artifacts unchanged rather than silently regenerating them, and the defect motivated the repairs in v1.4. The episode illustrates that structural validators for synthetic provenance benchmarks should check temporal consistency of lineage, not only counts and splits.

8 Reproducibility

Code, data, and logs are available at `https://github.com/theaayushdev/echotrace-benchmark`. The repository contains the deterministic generators, public/gold data separation, validators, detector code and validation-selected settings, the frozen confirmatory protocol, the study runner, raw append-only response logs including failures, and analysis scripts. Every table value in this paper is generated from analysis artifacts by `scripts/export_llm_study_results.py` and `scripts/export_v1_paper_results.py`; response counts quoted in the text come from `scripts/llm_count_breakdown.py`. The study requires no GPU and no paid API; it runs on CPU and free provider tiers. Raw logs contain no credentials.

9 Conclusion

In closed packets with visible provenance, one of three freely accessible models counted documents as independent sources, and no model's verdict confidence depended on source independence. Asking models to consider dependence removed overcounting but caused undercounting or abstention on genuinely independent evidence, even with correct provenance metadata. Provenance mitigations should therefore be judged on both directions of counting error. The next steps are the held-out run of the frozen protocol, blinded human review of the free-text answers, and real-source evaluation with retrieval.

Limitations

Synthetic construct. Documents are generated from templates and labels follow generator rules. In v1.4 provenance is usually visible through links and attribution, so the study measures whether models *use* visible provenance, not whether they discover hidden ancestry. Results on unseen prose styles reduce, but do not remove, the risk of template learning.

Exploratory evidence. All language-model results come from 12 validation worlds (8 complete worlds for the Groq models), which also informed the protocol revision. Intervals are wide, no multiplicity correction is applied, and no result should be read as a confirmed effect.

No human validation. No independent human annotation is complete. Primary outcomes are computed from a structured count, not from human judgments of whether free-text answers claim corroboration. The answers are retained for a later blinded human rubric.

Closed evidence packets. Models receive the four relevant documents. We do not evaluate retrieval, distractor rejection, or an autonomous agent. Connected components force non-overlapping families, whereas real articles may draw on several studies.

Models and access. We use three openly accessible models under free quotas. Provider aliases may change after the study date, and results may

not transfer to larger proprietary systems. Asking for $\widehat{K}$ may itself change behaviour; we treat it as part of the task definition, applied equally to all arms.

Responsible use. Dependence is not a proxy for truth, quality, or misconduct. Similar wording does not establish copying, and a shared family can be reliable. EchoTrace should not be used to accuse authors or rank publishers.

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